

ĐẠI HỌC BÁCH KHOA HÀ NỘI
TRƯỜNG CÔNG NGHỆ THÔNG TIN VÀ TRUYỀN THÔNG

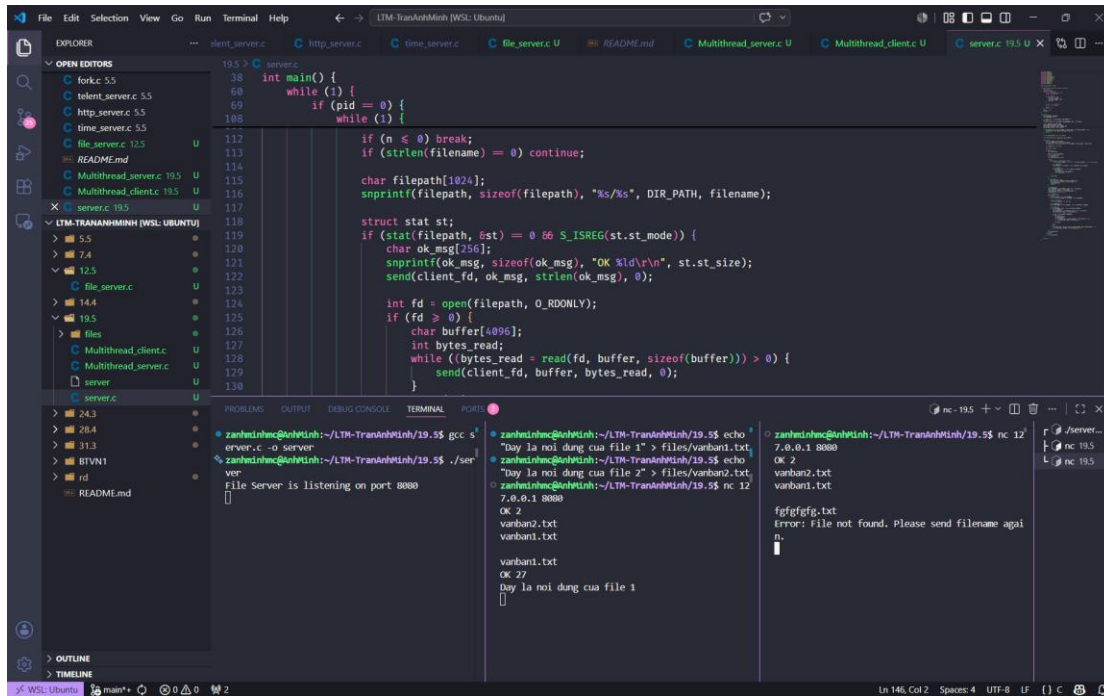
BÁO CÁO BÀI TẬP
LẬP TRÌNH MẠNG

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HÀ NỘI, 04/2026

Bài 3:

3.1



```
int main() {
    while (1) {
        if (pid == 0) {
            while (1) {
                if (n <= 0) break;
                if (strlen(filename) == 0) continue;

                char filepath[1024];
                snprintf(filepath, sizeof(filepath), "%s/%s", DIR_PATH, filename);

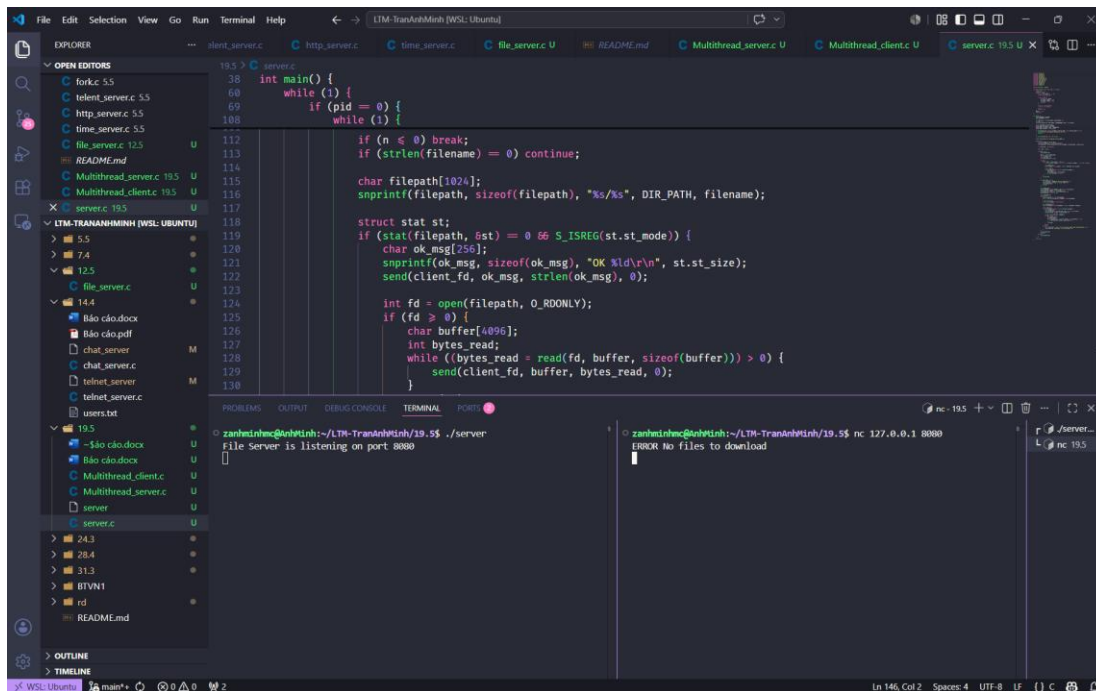
                struct stat st;
                if (stat(filepath, &st) == 0 && S_ISREG(st.st_mode)) {
                    char ok_msg[256];
                    snprintf(ok_msg, sizeof(ok_msg), "OK %ld\r\n", st.st_size);
                    send(client_fd, ok_msg, strlen(ok_msg), 0);

                    int fd = open(filepath, O_RDONLY);
                    if (fd >= 0) {
                        char buffer[4096];
                        int bytes_read;
                        while ((bytes_read = read(fd, buffer, sizeof(buffer))) > 0) {
                            send(client_fd, buffer, bytes_read, 0);
                        }
                    }
                }
            }
        }
    }
}
```

```
zanhminh@LTM-TranAnhMinh:~/LTM-TranAnhMinh/19.5$ gcc s
server.c -o server
zanhminh@LTM-TranAnhMinh:~/LTM-TranAnhMinh/19.5$ ./ser
File Server is listening on port 8080

nc - 19.5 +
r /server...
7.0.0.1 8080
OK 2
varban2.txt
varban1.txt
fgfgfgfg.txt
Error: File not found. Please send filename again.
r.
nc 19.5
```

Nếu có file



```
int main() {
    while (1) {
        if (pid == 0) {
            while (1) {
                if (n <= 0) break;
                if (strlen(filename) == 0) continue;

                char filepath[1024];
                snprintf(filepath, sizeof(filepath), "%s/%s", DIR_PATH, filename);

                struct stat st;
                if (stat(filepath, &st) == 0 && S_ISREG(st.st_mode)) {
                    char ok_msg[256];
                    snprintf(ok_msg, sizeof(ok_msg), "OK %ld\r\n", st.st_size);
                    send(client_fd, ok_msg, strlen(ok_msg), 0);

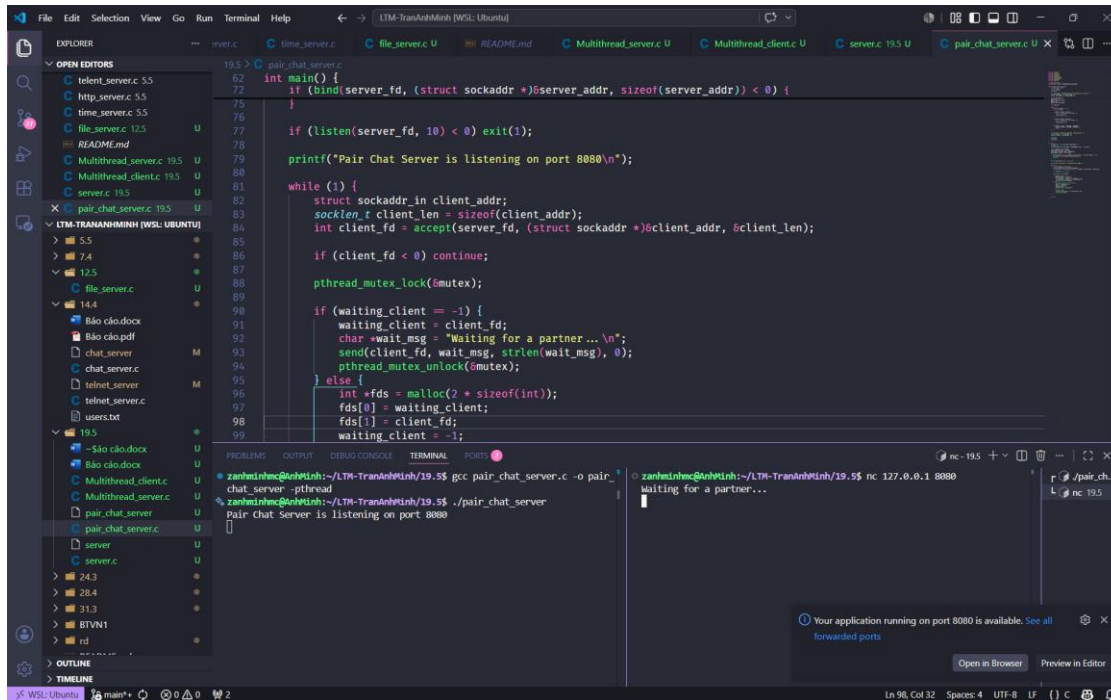
                    int fd = open(filepath, O_RDONLY);
                    if (fd >= 0) {
                        char buffer[4096];
                        int bytes_read;
                        while ((bytes_read = read(fd, buffer, sizeof(buffer))) > 0) {
                            send(client_fd, buffer, bytes_read, 0);
                        }
                    }
                }
            }
        }
    }
}
```

```
zanhminh@LTM-TranAnhMinh:~/LTM-TranAnhMinh/19.5$ ./server
File Server is listening on port 8080

nc - 19.5 +
r /server...
ERROR No files to download
r.
nc 19.5
```

Nếu không có file

3.2



```
int main() {
    if (bind(server_fd, (struct sockaddr *)&server_addr, sizeof(server_addr)) < 0) {
    }

    if (listen(server_fd, 10) < 0) exit(1);

    printf("Pair Chat Server is listening on port 8080\n");

    while (1) {
        struct sockaddr_in client_addr;
        socklen_t client_len = sizeof(client_addr);
        int client_fd = accept(server_fd, (struct sockaddr *)&client_addr, &client_len);

        if (client_fd < 0) continue;

        pthread_mutex_lock(&mutex);

        if (waiting_client == -1) {
            waiting_client = client_fd;
            char *wait_msg = "Waiting for a partner...\n";
            send(client_fd, wait_msg, strlen(wait_msg), 0);
            pthread_mutex_unlock(&mutex);
        } else {
            int *fds = malloc(2 * sizeof(int));
            fds[0] = waiting_client;
            fds[1] = client_fd;
            waiting_client = -1;
        }
    }
}
```

zanhminh@zanhminh:~/LTH-TranAnhMinh/19.5\$ gcc pair_chat_server.c -o pair_chat_server -pthread

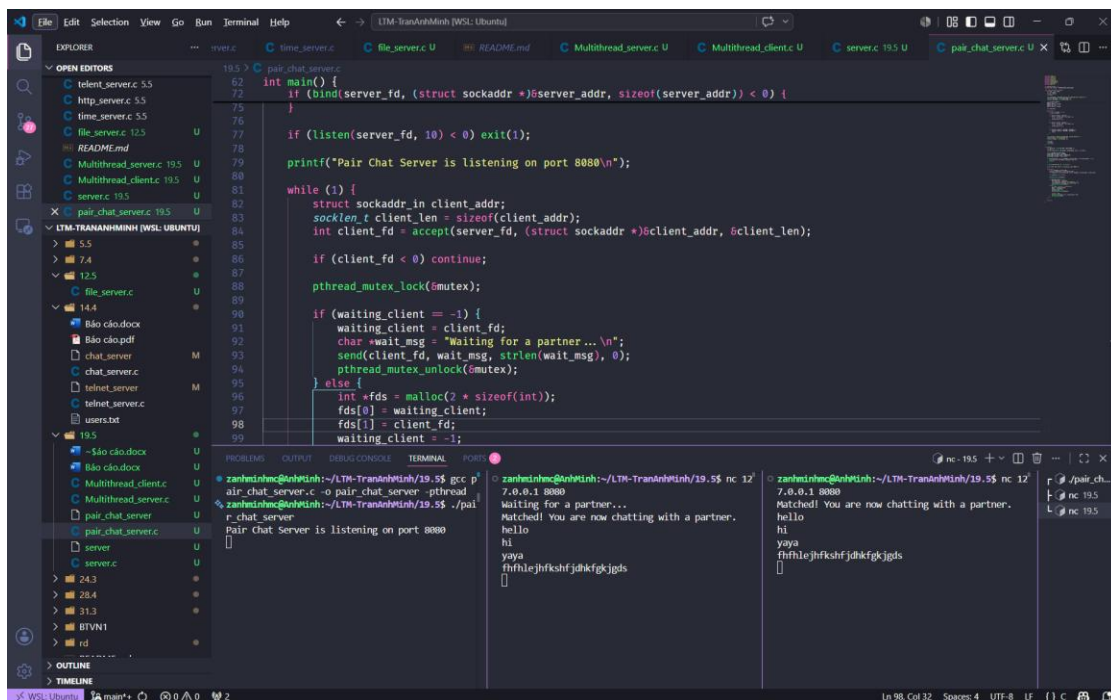
zanhminh@zanhminh:~/LTH-TranAnhMinh/19.5\$./pair_chat_server

Pair Chat Server is listening on port 8080

nc - 19.5

Waiting for a partner...

Khi chỉ có 1 client, client nhận thông báo chờ



```
int main() {
    if (bind(server_fd, (struct sockaddr *)&server_addr, sizeof(server_addr)) < 0) {
    }

    if (listen(server_fd, 10) < 0) exit(1);

    printf("Pair Chat Server is listening on port 8080\n");

    while (1) {
        struct sockaddr_in client_addr;
        socklen_t client_len = sizeof(client_addr);
        int client_fd = accept(server_fd, (struct sockaddr *)&client_addr, &client_len);

        if (client_fd < 0) continue;

        pthread_mutex_lock(&mutex);

        if (waiting_client == -1) {
            waiting_client = client_fd;
            char *wait_msg = "Waiting for a partner...\n";
            send(client_fd, wait_msg, strlen(wait_msg), 0);
            pthread_mutex_unlock(&mutex);
        } else {
            int *fds = malloc(2 * sizeof(int));
            fds[0] = waiting_client;
            fds[1] = client_fd;
            waiting_client = -1;
        }
    }
}
```

zanhminh@zanhminh:~/LTH-TranAnhMinh/19.5\$ gcc pair_chat_server.c -o pair_chat_server -pthread

zanhminh@zanhminh:~/LTH-TranAnhMinh/19.5\$./pair_chat_server

Pair Chat Server is listening on port 8080

nc - 19.5

Waiting for a partner...

nc - 19.5

Matched! You are now chatting with a partner.

hello

hi

yaya

thuhlejhfkshfjdahfkgkjgds

Khi có client 2 kết nối thì 2 client được kết nối và có thể chat

```
int main() {
    if (bind(server_fd, (struct sockaddr *)&server_addr, sizeof(server_addr)) < 0) {
    }

    if (listen(server_fd, 10) < 0) exit(1);

    printf("Pair Chat Server is listening on port 8080\n");

    while (1) {
        struct sockaddr_in client_addr;
        socklen_t client_len = sizeof(client_addr);
        int client_fd = accept(server_fd, (struct sockaddr *)&client_addr, &client_len);

        if (client_fd < 0) continue;

        pthread_mutex_lock(&mutex);

        if (waiting_client == -1) {
            waiting_client = client_fd;
            char *wait_msg = "Waiting for a partner...\n";
            send(client_fd, wait_msg, strlen(wait_msg), 0);
            pthread_mutex_unlock(&mutex);
        } else {
            int *fds = malloc(2 * sizeof(int));
            fds[0] = waiting_client;
            fds[1] = client_fd;
            waiting_client = -1;
        }
    }
}
```

Terminal output:

```
zanhseinhc@LTM-TranAnhMinh:~/LTM-TranAnhMinh/19.5$ gcc p
zanhseinhc@LTM-TranAnhMinh:~/LTM-TranAnhMinh/19.5$ ./pai
Pair Chat Server is listening on port 8080
Waiting for a partner...
Matched! You are now chatting with a partner.
hello
hi
yaya
fhhlejhfkshfjdskfkgjds
Partner disconnected. Closing session.
```

Khi 1 client ngắt kết nối, đóng kết nối với client còn lại

Bài 4

Chat server

```
int main() {
    int server_fd = socket(AF_INET, SOCK_STREAM, 0);
    int opt = 1;
    setsockopt(server_fd, SOL_SOCKET, SO_REUSEADDR, &opt, sizeof(opt));

    struct sockaddr_in server_addr;
    server_addr.sin_family = AF_INET;
    server_addr.sin_addr.s_addr = INADDR_ANY;
    server_addr.sin_port = htons(8080);

    if (bind(server_fd, (struct sockaddr *)&server_addr, sizeof(server_addr)) < 0) {
        printf("Error: Port 8080 is already in use.\n");
        exit(1);
    }

    if (listen(server_fd, 10) < 0) exit(1);

    printf("Chat Server (Multithreaded) listening on port 8080\n");

    while (1) {
        struct sockaddr_in client_addr;
        socklen_t client_len = sizeof(client_addr);

        int *client_fd = malloc(sizeof(int));
        *client_fd = accept(server_fd, (struct sockaddr *)&client_addr, &client_len);
    }
}
```

Terminal output:

```
zanhseinhc@LTM-TranAnhMinh:~/LTM-TranAnhMinh/19.5$ gcc c
zanhseinhc@LTM-TranAnhMinh:~/LTM-TranAnhMinh/19.5$ ./cha
Chat Server (Multithreaded) listening on port 80
Please enter client_id: client_id: c1
Accepted
2026/05/24 02:14:24PM C1: sdfsdgs
2026/05/24 02:14:25PM C1: asdaf agg
```

telnet server

```
File Edit Selection View Go Run Terminal Help
LTM-TranAnhMinh [WSL: Ubuntu]

EXPLORER
  OPEN EDITORS
    time_server.c 5.5
    file_server.c 12.5
    Multithread_server.c 19.5
    Multithread_client.c 19.5
    server.c 19.5
    pair_chat_server.c 19.5
    chat_server.c 19.5
    telnet_server.c 19.5
    database.txt 19.5
  LTM-TRANANHMINH [WSL: UBUNTU]
    5.5
    7.4
    12.5
    file_server.c
    14.4
    19.5
    -Sao cao.docx
    Bao cao.docx
    chat_server
    chat_server.c
    database.txt
    Multithread_client.c
    Multithread_server.c
    pair_chat_server
    pair_chat_server.c
    server
    server.c
    telnet_server
    telnet_server.c
    24.3
    28.4
    31.3
    BTVN1
    rd
    README.md
  OUTLINE
  TIMELINE

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
zanhminh@LTM-TranAnhMinh:~/LTM-TranAnhMinh/19.5$ gcc telnet_server.c -o telnet_s
zanhminh@LTM-TranAnhMinh:~/LTM-TranAnhMinh/19.5$ ./telnet_server
telnet Server (Multithreaded) listening on port 8080
zanhminh@LTM-TranAnhMinh:~/LTM-TranAnhMinh/19.5$ nc 127.0.0.1 8080
Username: gggfgh
Password: yfhghs
Login failed. Try again.
Username: admin
Password: gghhhh
Login failed. Try again.
Username: admin
Password: admin
Login successful.
> dir
Bao\ cao.docx      chat_server.c      pair_chat_server.c telnet_server.c
Multithread_client.c database.txt      server      -Sao\ cao.docx
Multithread_server.c out_4.txt      server.c
chat_server      pair_chat_server telnet_server
>
```

Time server

```
EXPLORER
  OPEN EDITORS
    time_server.c 19.5
    file_server.c 12.5
    Multithread_server.c 19.5
    Multithread_client.c 19.5
    server.c 19.5
    pair_chat_server.c 19.5
    chat_server.c 19.5
    telnet_server.c 19.5
    database.txt 19.5
  LTM-TRANANHMINH [WSL: UBUNTU]
    5.5
    7.4
    12.5
    file_server.c
    14.4
    19.5
    -Sao cao.docx
    Bao cao.docx
    chat_server
    chat_server.c
    database.txt
    Multithread_client.c
    Multithread_server.c
    pair_chat_server
    pair_chat_server.c
    server
    server.c
    telnet_server
    telnet_server.c
    time_server
    time_server.c
    24.3
    28.4
    31.3
    BTVN1
    rd
    README.md
  OUTLINE
  TIMELINE

int main() {
    int server_fd = socket(AF_INET, SOCK_STREAM, 0);
    int opt = 1;
    setsockopt(server_fd, SOL_SOCKET, SO_REUSEADDR, &opt, sizeof(opt));

    struct sockaddr_in server_addr;
    server_addr.sin_family = AF_INET;
    server_addr.sin_addr.s_addr = INADDR_ANY;
    server_addr.sin_port = htons(8080);

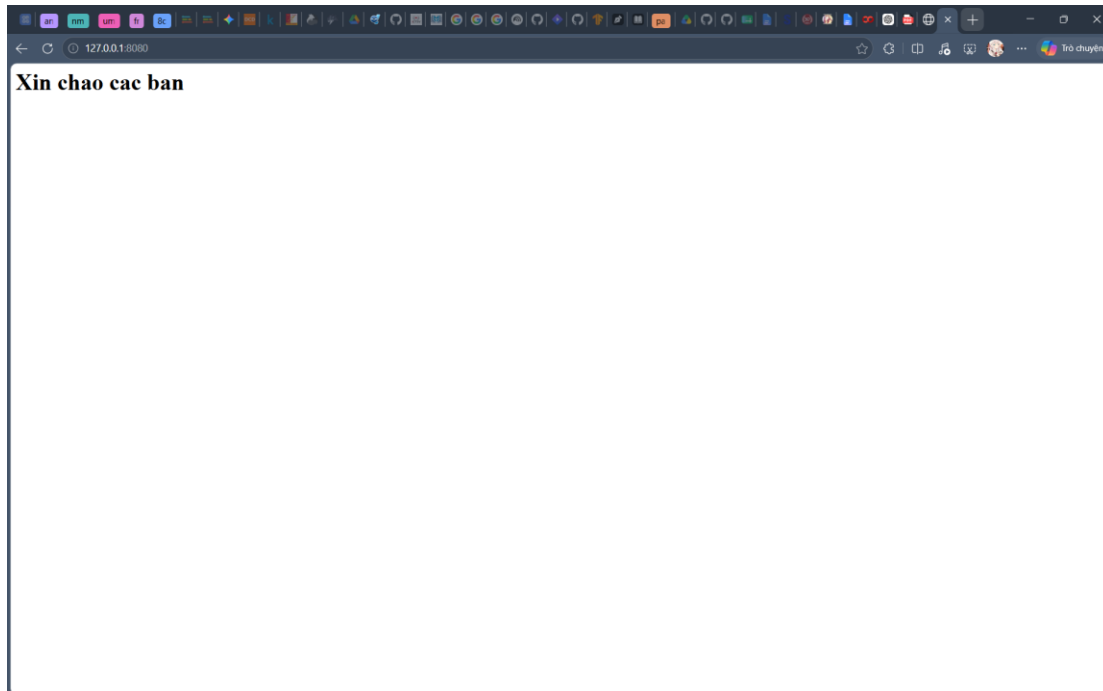
    if (bind(server_fd, (struct sockaddr *)&server_addr, sizeof(server_addr)) < 0) {
        printf("Error: Port 8080 is already in use.\n");
        exit(1);
    }

    if (listen(server_fd, 10) < 0) exit(1);

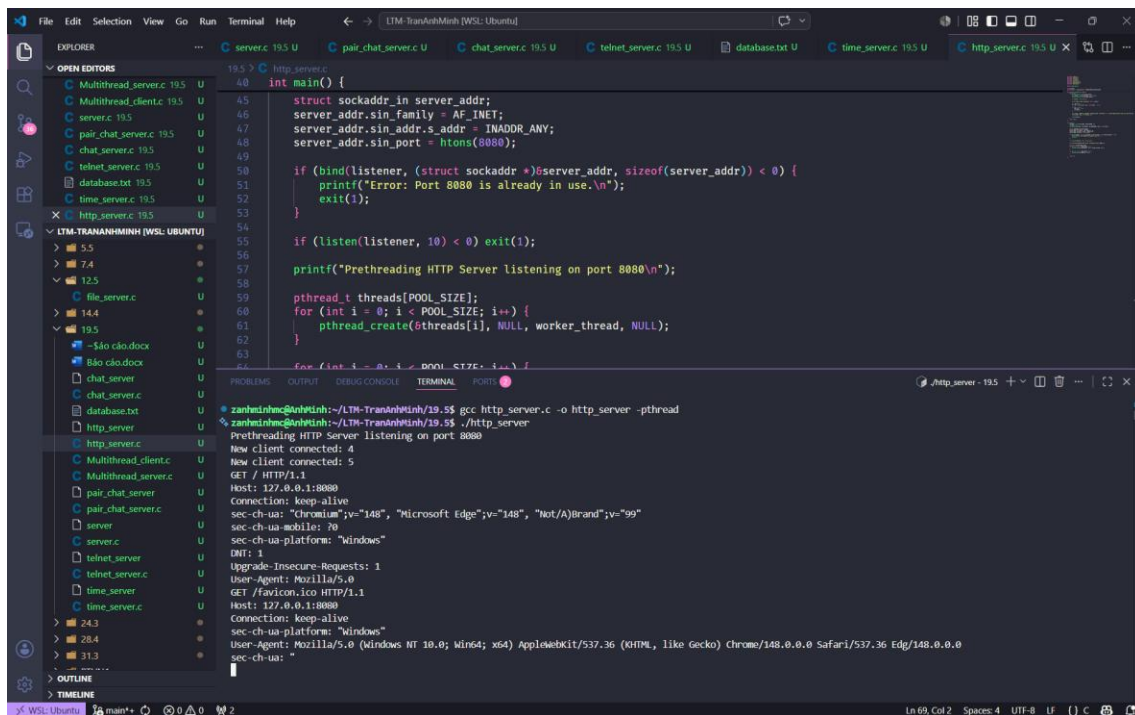
    printf("Time Server (Multithreaded) is listening on port 8080\n");

    GET_TIME dd/mm/yyyy
    24/05/2020
    GET_TIME mm/dd/yy
    05/24/26
    GET_TIME dd/mm/yy
    24/05/26
    GET_TIME mm/dd/yyyy
    05/24/2020
    GET_TIME hh/mm/ss
    Error: Unsupported time format
    }
```

Http server



Trang web mở ra



Server báo có kết nối

